# Pulock Das

Software Engineer | AI/ML Engineer

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## Summary

Software engineer seeking **Software Engineer** and **AI/ML Engineer** roles. Brings 1.5 years of production backend experience in fintech, building transaction services, asynchronous messaging, and database performance work at scale, plus graduate research in deep learning for medical imaging, federated learning, and efficient Vision Transformers. Particularly interested in distributed system design and in putting machine learning models into reliable production systems.

## Education

**Tennessee State University** **Aug 2025 – Dec 2026 (Expected)**

*M.S. in Computer & Information Systems Engineering · GPA 4.00/4.00* *Nashville, TN*

- **Coursework:** Software Systems Design, Digital Image Processing

**Shahjalal University of Science and Technology** **Feb 2018 – May 2023**

*B.Sc. (Eng.) in Software Engineering · CGPA 3.62/4.00* *Sylhet, Bangladesh*

- **Coursework:** Data Structures & Algorithms, Design Patterns, Relational Database Management Systems

## Technical Skills

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|---|---|
| **Languages** | Java, Python, SQL, C++, Bash |
| **Backend** | Spring Boot, Spring MVC, Spring AMQP, Django, Django REST Framework, REST APIs |
| **Data & Messaging** | PostgreSQL, Redis, RabbitMQ |
| **Machine Learning** | PyTorch, scikit-learn, NumPy, pandas, Flower; CNNs, Vision Transformers, federated learning, deep clustering |
| **DevOps & Tools** | Docker, Linux, Git, Jenkins, GitLab CI/CD, JMeter |

## Professional Experience

**Software Engineer** **Feb 2023 – Aug 2024**

*Tallykhata* *Dhaka, Bangladesh*

- Built and maintained **Java/Spring Boot** and **Django** APIs for high-volume financial workflows and integrated new payment methods.
- Moved transaction processing to an asynchronous producer–consumer design with **Spring AMQP** and **RabbitMQ**; clustered broker nodes for high availability.
- Batched calls to external notification services to reduce outbound request volume.
- Tuned complex **PostgreSQL** queries using execution plans (EXPLAIN/ANALYZE), and added **HikariCP** connection pooling to reduce API latency.
- Enforced data integrity for financial records with database triggers and constraints.
- Developed APIs on the platform's **OAuth**, **JWT**, and session-based authentication and authorization flows.
- Set up integration tests and **CI/CD** pipelines with Jenkins and GitLab; diagnosed production issues using service logs and monitoring.

**Software Engineering Intern** **Sep 2021 – Jun 2022**

*Progoti Systems Ltd.* *Dhaka, Bangladesh*

- Built and validated REST endpoints for Tallypay credit collection workflows using **Django REST Framework** and PostgreSQL.
- Implemented API throttling with **Redis** to protect public endpoints from abuse.
- Worked on asynchronous processing with RabbitMQ and Spring AMQP; load-tested services with **JMeter** in a Scrum team.

## Research Experience

**Graduate Research Assistant** **Aug 2025 – Present**

*Tennessee State University* *Nashville, TN*

- Developed **PhysNet**, a **PyTorch** model that embeds tumor growth dynamics into **ResNet-50** features for medical image classification; published at IEEE SoutheastCon 2026.
- Evaluated federated medical imaging models with **FedAvg** and secure aggregation, comparing accuracy, macro-F1, and calibration across simulated clients.
- Benchmarked deep clustering and tabular foundation models (TabPFN) on **16 imbalanced datasets** and developed an ensemble method for unsupervised clustering.
- Explored token reduction and knowledge distillation for efficient, encrypted **Vision Transformer** inference.

## Projects

**Federated Medical Image Classification** | *PyTorch, Flower* GitHub

- Trained and evaluated physics-informed models on brain MRI, colon histopathology, and chest X-ray datasets in a federated setting.

**Peer Review Evaluation System** | *Django, PostgreSQL, Agile* GitHub

- Built a peer review and evaluation web application with Django and PostgreSQL, developed in Agile sprints as a Tennessee State University Software Systems Design project.

**Intersection Safety Data Analysis** | *Python* GitHub

- Conducted exploratory data analysis on intersection safety data.

**File Transfer System** | *Java, Sockets, Swing*

- Built a client–server desktop application for interactive file transfers over Java sockets.

## Publications


[1] **P. Das**, A. Amin, K. Hasan, R. Thompson, A. D. Okpalaeze, L. Hong. Physics-Embedded Feature Learning for AI in Medical Imaging. *IEEE SoutheastCon*, 2026.

[2] **P. Das**, Y. Hou, M. K. Bhuiyan, M. D. Samad. Ensemble of Unsupervised Deep Learning for Clustering Imbalanced Tabular Data. *Submitted to ICECE 2026*; arXiv:2608.00346.

[3] M. Mia, **P. Das**, A. Habib. Verse-Based Emotion Analysis of Bengali Music from Lyrics Using Machine Learning and Neural Network Classifiers. *International Journal of Computing and Digital Systems*, 2024.


## Honors & Certifications

**2nd Place, Graduate Poster in Engineering** | *TSU University-Wide Research Symposium* 2026

**Neural Networks and Deep Learning** | *DeepLearning.AI, Coursera*

**LinkedIn Skill Badges** | *Data Structures; Object-Oriented Programming*

**Competitive Programming** | *850+ problems solved across LeetCode, Codeforces, VJudge, StopStalk*